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DECEMBER 35 CENTS

SCIENCE FICTION

IN THIS ISSUE! A thrilling new short novel by

ROBERT SILVERBERG

An unusual science-espionage story by

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*** START OF THE PROJECT GUTENBERG EBOOK EDDIE ***

EDDIE

BY FRANK RILEY

*It's no surprise that the top brass was
in a complete swivet; Eddie knew answers to
questions that weren't even asked. What's
more, nothing was a secret with him around!*

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Philip Duncan, the St. Louis attorney and former FBI agent, who wrote the definitive "History of Espionage", observes that in all the records dealing with spies and counterspies there is no more significant case than that of Dr. John O'Hara Smith, an electronics research engineer. Duncan maintains that Dr. Smith, whose rather quixotic name is real and not assumed, contributed more to the advancement of espionage and counter-espionage methods than any one person in history.

For a period of more than a year, the case of Dr. John O'Hara Smith was known to only a few security and defense officials. The first public reference to it came on November 22, 1956, when an assistant to Secretary of Defense Wilson obliquely commented on it in testimony before the House Military Affairs Committee. Subsequently, more details were leaked to several Washington correspondents, and then vigorously denied. A brief account of the matter appeared on an inside page of the New York Times, but aroused no general interest.

As a matter of fact, so little is known about the entire case that several of the people who were in on its early phases are still not sure whether Dr. John O'Hara Smith is alive or dead, or whether he was a spy or counterspy.

However, on the basis of information now declassified, plus two highly technical papers presented to the Institute of Research Engineers, anyone sufficiently interested can reconstruct most of the case.

It began at approximately 7:15 P.M., August 11, 1955, when Dr. John O'Hara Smith returned with a bag of groceries to his house trailer in the Mira Mar Trailer Park, overlooking a long blue reach of the Pacific Ocean, some twelve miles south of Los Angeles. He put the groceries on the drainboard beside his spotless two-burner butane stove, carefully flicked away a speck of dust and then stepped eagerly toward the rear of his trailer, where an intricate assembly of tubes and wires occupied what normally would have been the dining area.

Dr. Smith flipped on a switch, and then received what he later called, in his precise, pedantic way, a split-second premonition of danger.

The Go-NoGo panel light flashed and went out; the transistor looked grey instead of red; the wires to the binary-coded digitizer were crossed; the extra module in the basic assembly had not been there that morning....

Dr. Smith methodically catalogued these details, and he stepped backward, just a breath of a moment before the low hum sharpened to a whine. He tripped, and in falling his left shoulder knocked open the door to the small toilet closet. Instinctively, he writhed the upper part of his body through the narrow doorway. His thick-lensed glasses fell underneath him, leaving him practically blind.



His elbows and knees were still making frenzied, primordial crawling movements when the detonation brought a wave of oblivion that almost, but not quite, preceded the pain.

A squad car from the Los Angeles County Sheriff's Department turned in the first report:

John O'Hara Smith, male, white, about 45; critically injured by explosion in house trailer; removed by ambulance to General Hospital; explosion occurred at....

Two days later, the Sheriffs Department apparently closed the case with a one-line addition to its original report:

Explosion believed to have been caused by leaking butane connection.

But, in the interval, other agencies had entered the case.

The first was the Industrial Security Office attached to the Western Division of the Air Force's Research and Development Command in the once suburban community of Inglewood, California.

When Chief Security Officer Amos Busch received a call at 11:32 the morning after the explosion, he automatically noted the time on his desk pad. The call was from Pacific Electronics, Inc., a subcontracting firm in nearby El Segundo.

The president and owner of Pacific Electronics was on the phone. In a tone that betrayed considerable agitation, he identified himself as Wesley Browne.

"One of my research engineers—my best engineer, dammit—was nearly killed last night in an explosion ... maybe he's dead now," reported Browne, his words breathlessly treading on each other. "There's something damn funny about this...."

Amos Busch wrote: Research engineer ... explosion ... nearly killed. Then he asked judicially:

"What do you mean by 'damn funny', Mr. Browne?"

"This engineer was working on our vernier actuating cylinder for the Atlas guided missile.... Just two days ago, he—he said he wanted me to know where his files were ... in case anything happened to him...."

Amos Busch was a jowly, greying man who gave the appearance of being slow moving. But before the president of Pacific Electronics, Inc., hung up, Busch had already used another phone and the intercom to put in motion a chain reaction that would deliver to his desk the security report on Dr. John O'Hara Smith.

There was nothing out of order in the report. There couldn't have been, or Dr. Smith wouldn't have been cleared for the ballistic missile program. According to the report, he had lived aloofly for all of his adult years. Even as a boy, his sole interest had been to tinker with mechanical projects. His grades and IQ were high above the norm, and his attitude towards his classmates varied between impatience and out-right sarcasm. "I always thought John was a lonely boy," a former teacher had recalled to an FBI officer during the security check. "He never had anything in common with other youngsters." After obtaining his Ph.D. in electrical engineering from the University of Wisconsin, he had worked for Allis-Chalmers Research Division in Milwaukee and lived with his mother until her death in 1951, when he bought a house trailer and moved to the coast. He had no close friends, no record of even a remote connection with any communist or communist-front group.

Security Officer Busch decided to visit the trailer, or what remained of it. He was not an electronics man, or even a normally incompetent do-it-yourself mechanic, but when he saw the shattered tangle of wires and tubes, along with the obvious remnant of a short-wave receiver, Amos Busch promptly called Major General David Sanders, commander of the USAF's Western Development Division.

General Sanders scratched his tanned bald head, and said,

"We'd better get the FBI in on this, Amos."

The FBI went to work with a thoroughness that made John O'Hara Smith's previous security investigation look like the processing of an application to join the Kiwanis. While agents sifted every detail of his life since the day of

his birth, he was moved to a private room at General Hospital and three nurses cleared for security were assigned to care for him.

For eight days, Smith was in a coma. On the morning of the ninth day, he groaned, turned to one side and rolled back again. The nurse on duty put down her magazine and moved quickly to his bedside. She moistened a cloth and wiped the perspiration from his high forehead, brushing back the thinning tangle of fine, brown hair.

His eyes blinked open, stared at her. He whispered:

"Eddie ... what happened ... to Eddie?"

Remembering her instructions from the FBI, the nurse turned to make certain the door was closed.

"Was Eddie in the trailer with you?" she asked, bending closer to catch his reply.

He gave her a look of utter disgust, and tried to moisten his cracked lips with the tip of his tongue. But he drifted off again without replying.

This incident was duly recorded in the FBI's growing dossier, along with another conversation that took place in the office of Wesley Browne at Pacific Electronics, Inc. After carefully reviewing John O'Hara Smith's work record, FBI agent Frank Cowles inquired:

"Is there anything—anything at all, Mr. Browne—that you would consider out of the ordinary about Smith's recent actions?"

There was a trace of uneasiness in Browne's manner, but he tried to cover it by looking annoyed.

"I don't know why in the devil you fellows are spending so much time on Smith!... He sure as hell didn't blow himself up!"

"Of course not," Cowles said, placatingly. "But we never know where a lead will come from...."

He repeated the question.

Browne hesitated.

"I suppose," he began, shifting his big bulk uncomfortably, "this will sound kind of odd ... but you know we've got the subcontract to produce this

actuating cylinder for the Atlas...."

The agent nodded.

"Well, six months before we were asked to submit specs and bids on such a cylinder, Smith came to me and said he had an idea for something the Air Force might soon be needing...."

Agent Cowles maintained his air of polite attention, but his cool grey eyes narrowed. Browne shifted again, and continued:

"I told him to go ahead—you never can tell what these research guys will come up with...."

"And what did he come up with, Mr. Browne?"

"You won't believe this, maybe—but he came up with the design for the complete vernier hydraulic actuating cylinder—including the drive sector gear—at least three months before we had the faintest idea such an item would even be needed!"

The FBI man's ball-point pen moved swiftly.

"Anything else?"

Browne instinctively lowered his voice:

"Smith even suggested that the cylinder would help to offset the roll and yaw in an intercontinental ballistic missile!"

A brittle edge came into the agent's courteous tone:

"Did you report this to security?"

In spite of the air-conditioning unit in the window, the president and owner of Pacific Electronics, Inc., seemed to feel that the room was getting very warm. He ran a fat forefinger under his white collar.

"No," he admitted. "We got the contract, of course—it was a cinch!—and I just wrote it off as a lucky break.... You can see how I'd feel, can't you?"

"Yes," said Cowles, "I can."

Bit by bit, a new picture of the meticulous, professorial Dr. Smith began to emerge from the FBI dossier.

During the working week, his habit had been to keep his trailer in a small park just off Sepulveda Blvd., a half-mile from the Pacific Electronics plant. After work on Fridays, he invariably left for the weekend, usually for any one of a dozen scenic trailer parks along the coast between San Diego and Santa Barbara. He always went alone. No one had ever seen or met "Eddie". Outside of working hours, Smith's only association with his professional colleagues was through the Institute of Research Engineers. He attended monthly meetings, and occasionally wrote dry, abstract articles on theoretical research for the Institute's quarterly journal.

Under microscopic study and chemical analysis, investigators determined that nitro-glycerine had caused the explosion. The fused mass of electronics wreckage in Smith's trailer were identified as parts of a computer assembly. Thousands of dollars had been spent on components over the past three years. Purchases, usually for cash, were traced to various electronic supply companies in the greater Los Angeles area.

Dr. Smith's bank account showed a balance of only \$263.15. But the big find came from a safety deposit box in the same branch bank. There, along with a birth certificate, his mother's marriage license, an insurance policy, his doctor's degree from the University of Wisconsin and an unused passport, was a duplicate set of computer memory tapes.

It took the FBI forty-eight hours to play a few selected segments from these tapes, which obviously had been recorded over a period of several years.

Two notations made by Agent Cowles indicate the type of material contained on the tapes:

"If a deliberate attempt were made to run a thermonuclear test explosion within the frontiers of Russia, in such a way as to avoid detection, it would almost certainly be successful...."

"The Soviet Union may soon develop a new ratio of fusion to fission energy in high yield weapons and will require additional data...."

FBI agents listening to these playbacks were convinced, almost to a man, that they had stumbled across the hottest espionage trail since the arrest of

Klaus Fuchs and the case of the Rosenbergs.

A round-the-clock security guard was placed outside the hospital room of John O'Hara Smith, while Federal authorities waited impatiently to see whether he would live or die. Smith would answer, or leave unanswered, a lot of vital questions.

Security notwithstanding, it was the day after Labor Day before the medical staff of General Hospital would permit the first direct questioning of Dr. Smith. And then the interrogators were instructed:

"Only a few minutes."

Three men filed quietly into Smith's room as soon as the nurse removed his luncheon tray. They stood in a semi-circle around the foot of his bed.

Agent Frank Cowles opened a black leather folder the size of a small billfold and presented his credentials. He introduced General Sanders and Security Officer Busch. It was the first time any of the men had seen John O'Hara Smith. The reports had called him pudgy, but now he had lost twenty pounds and his cheek bones were gaunt under his pallid skin. He wore unusually thick, dark-rimmed glasses that magnified his eyes and gave him an owlsh appearance. He returned their scrutiny with a mixture of assurance and impatience, like a professor waiting for his class to come to order.

"Good morning, gentlemen," he said tartly. "It's about time someone came to see me about this...."

Cowles cleared his throat and suggested cautiously:

"Then you're willing to give us a statement, Dr. Smith?"

"Don't talk drivels, man! How are you going to know anything about it if I don't make a statement!"

Though still weak, Dr. Smith's voice had a high, imperious quality. Clearly, he did not wish to waste time or strength on mere conversation.

The three men exchanged glances. Cowles and Amos Busch took out notebooks.

"Now, Dr. Smith," Cowles began, "what is your view as to the nature of the explosion in your trailer and the reason for it?"

"I'm an electronics research engineer, not an expert in explosives," Smith retorted with some asperity. "But as to the reason, I'm sure they wanted to destroy Eddie and me!"

He glared, as if daring anybody to challenge this statement.

"Eddie?" ventured Cowles.

"I try to speak plainly, Mr. Cowles.... I said 'Eddie and me'!"

General David Sanders rested two large hands on the foot of the white iron bedstead and squeezed until his knuckles bulged ominously. A volatile man, he had trouble with his own temper even without being provoked. But his voice was deceptively calm:

"Dr. Smith, do I gather that someone else was in the trailer with you at the time of the explosion?"

Smith grimaced expressively, and answered as if speaking to an eight-year-old:

"No, General Sanders.... I was quite alone."

After thirty years in the Air Force, Amos Busch was not used to hearing a Major General spoken to in this way. It violated his sense of propriety.

"Dr. Smith," he exploded, "just who or what in the hell is or was Eddie?"

With what was remarkably close to an air of incredulity, Smith looked slowly from one to the other.

"I gather you gentlemen haven't read my latest article."

"Not thoroughly," Cowles admitted.

"Then you don't know of my research work with an educatable computer," Smith said accusingly. Seeing that they didn't, he added: "I have named it 'Eddie'!"

"What ... what is an educatable computer?" ventured Cowles.

It was clear that Dr. Smith welcomed this question. His eyes glowed behind their thick lenses, and his high voice dropped its edge of sharpness.

"Eddie is a computer with a capacity to learn," he replied proudly. "It learns from assimilation of information and deductive reasoning—at a rate at least 10,000 times that of the human mind! That's why Eddie comes up with so many answers!... The only problem is, we seldom know what questions the answers answer."

His three interrogators had the look of men leaning into a heavy wind. General Sanders recovered first, and demanded:

"What the devil was it made for then?"

"Eddie was not designed for any specific task—that's why Eddie is so valuable ... and dangerous!"

Dr. Smith rolled out this last word as if he relished it.

"Do you realize," he went on, with careful emphasis, "that Eddie has solved problems we won't even know exist for another thousand years!"

This pronouncement was greeted by a moment of strained silence. General Sanders finally said,

"H-m-m-m."

He looked at Busch, who looked at Cowles, who asked:

"Does Eddie solve any problems closer to our own time, Dr. Smith?"

"Of course...."

"Did Eddie come up with the idea for that Atlas stabilizing cylinder?"

"Certainly."

General Sanders moved a step closer to the bed.

"Any other ideas like that?" he inquired eagerly.

Dr. Smith's smile was neither wholly supercilious nor merely self-assured. It was a little of both, plus a lot of pure satisfaction at being stage center with his favorite subject. He cocked his head back and stared down his stump of a nose.

"You're working on a missile defense system for bombers, aren't you?" he challenged General Sanders.

"What about it?" hedged the General.

"Have you learned how to design a finned missile which can be launched across the bomber's airstream without being thrown off course?"

General Sanders ignored a warning glance from Amos Busch.

"Do you ... does this Eddie know how to do it?"

"Eddie says it doesn't matter!"

"What?"

"Eddie says what difference does it make if the missile is thrown off course by the airstream—as long as you can reorient it into a compensated trajectory. We were working on a new gyroscope principle that might do the trick...."

FBI Agent Cowles was always the personification of courtesy, but he could assert himself when necessary. He did so now.

"Excuse me, General," he interrupted, "but first there are some other matters we must go into with Dr. Smith."

The General nodded reluctantly. He took out an envelope and made some notes of his own on the back of it.

"Now, Dr. Smith," said Cowles, "let's get back to the explosion.... Why do you feel someone wanted to destroy you and Eddie?"

"I believe they had copied Eddie's circuit design and wanted to make sure another one wasn't built—at least in the immediate future."

"Why not?"

Dr. John O'Hara Smith showed a neat flair for timing as he waited just long enough to build suspense, before answering:

"Because Eddie knew that our security system for safeguarding the missile program is about as up to date as the horse and buggy!"

His words couldn't have been better chosen to startle his audience. Amos Busch took them as a personal affront.

"Horse and buggy!" he snorted. "You'd better spell that out, Dr. Smith!"

Smith's reply was prompt and precise:

"Eddie has concluded that human methods and minds alone are not enough to cope with security issues in an area where even the simplest technical problems must be handled by intricate computing devices...."

His owlsh eyes moved from one man to another, trying to judge whether they were following him.

"You see, Gentlemen," he went on, "the technology we are dealing with is so unbelievably complex that the possibilities for espionage are multiplied infinitely beyond the capacity of a human intellect to grasp and evaluate...."

"For example," demanded General Sanders.

"For example," Smith retorted with equal sharpness, "what good does it do to surround ballistic missile plants with security regulations if the missile itself can be stolen right out of the air?"

"Fantastic!" said General Sanders.

"Nuts," said Amos Busch.

Agent Cowles said nothing.

John O'Hara Smith sank back against his pillow, panting a little. His high forehead glistened with sweat. When he gathered the strength to speak again, he directed his words to General Sanders:

"General, these ICBM missiles being fired into the Atlantic Ocean from the coast of Florida.... Are you sure you know what's happened to all of them?"

"I think so," the General answered calmly.

"And what about your own X-15 project, General?"

The question was almost a taunt.

General David William Sanders had jumped with his paratroopers into France on a morning in June, 1944. He had risen in rank through the test of battle and the more excruciating ordeal of the Pentagon. He was a rock-jawed, six-foot, two-hundred pound man whom little could shock and

nothing could deter. But he had never faced a challenge like the seconds of silence that followed Dr. Smith's mocking question.

There was nothing he dared say, yet in saying nothing he was saying everything. FBI Agent Frank Cowles looked at him, then looked quickly away. Security Officer Busch studied his own hands as though discovering them for the first time.

The tableau remained frozen and silent until the door opened and a doctor said,

"That's all for today, gentlemen."

The three men left without a word.

Dr. John O'Hara Smith closed his eyes. On his pale lips was the suggestion of a smile.

When they were alone in the General's staff car, Amos Busch exhaled and said,

"I'll be damned."

"I gather," observed Cowles drily, "that something called an X-15 has turned up missing."

"A week ago," sighed General Sanders. "Somewhere in the Mojave Desert near Lancaster.... It was a very elementary prototype—the actual X-15 won't be ready for another three years...."

"Any idea what happened to it?"

"It was on a routine test flight and ran out of the tracking screen—headed northwest.... We haven't found a splinter from it! But there's a lot of rough country around there."

"Who knows it was lost?"

"Just the local base and our headquarters staff. The Pentagon, too, of course."

"And Dr. Smith," added Amos Busch, incredulously.

The staff car detoured off the freeway to deliver Cowles to the Federal Building.

"What do you make of this, Frank?" the General asked him.

"I'm just supposed to be gathering information."

"Oh, hell! We've been talking and you've been thinking—what?"

Cowles grinned.

"I've been thinking how lucky it is I don't have to make a decision about Smith!"

"So?"

"So we'll question him again tomorrow.... As long as he's willing to talk, the more he says, the better."

But, next morning, the medical staff again exercised its veto power. John O'Hara Smith had developed an infection and fever during the night. There could be no further questioning for the time being.

On the second day, when his fever ebbed, Dr. Smith irascibly ordered a pad of paper and began an interminable series of sketches. The nurse managed to sneak out a few of them, and FBI experts sat up all night vainly trying to figure out what they meant.

The following evening, when the last visitor's bell had sounded and the patients were bedded down for the night, Dr. Smith was staring unblinkingly into the dark shadows of his room. He had been given a sleeping pill at 9:30, but had held it under his tongue until the nurse left, and then had put it on the night table behind his thick-rimmed glasses. He seemed to be struggling with a problem. Once he turned on the night light, put on his glasses and made several rapid sketches that vaguely resembled a spider web.

A half hour later, his eyes began to droop. He picked up the sleeping pill, rolled it between his thumb and forefinger, then put it back on the table. His breathing became deeper.

A sound startled him awake. It was an odd sound, not a part of the subdued hospital noises. It was a persistent, metallic, scraping sound, and it came from outside his window.

Dr. John O'Hara Smith grabbed his glasses and rolled out of bed. He bunched up his pillow under the covers and crawled into the deeper darkness of the corner to the left of the window, which was open several inches. He crouched there, knees quivering from weakness.

There followed an interval of almost inaudible prying at the screen, broken by periods of silence as someone outside the second-story window apparently paused to listen. Finally, the screen was released with a faint pop. The lower half of the double-hung window eased upwards.

Again there was silence, save for the distant clatter of the self-service elevator.

Abruptly, a pencil-thin beam of light shot through the room, toward the bed. It focussed on the mound made by the pillow.

Short tongues of flame leaped out three times, with soft, spitting sounds. The pillow and the tangle of blankets twitched realistically.

The beam of light winked out; the screen plopped back into place. There were a few hasty, sliding noises of retreat, and that was all.

John O'Hara Smith's breath came in short, strained gasps, as though he were choked up with asthma. When he got control of himself, he eased back the edge of the drape and looked out the window. It was nearly twenty feet to the ground. A car turned off the boulevard, and came up the side street. The glow of its headlights briefly silhouetted the ladder angled against the side of the hospital.

Dr. Smith sat on the edge of the bed to think things over. His left thumb probed the holes in the blanket and pillow. This seemed to make up his mind.

He got his clothes from the closet and dressed as quickly as he could force his hands to move and co-ordinate. His trousers hung so loosely that the last hole in his belt made no difference. He pulled the belt tight and knotted it.

Next, he carefully folded his sketches and put them in the inside pocket of his coat. As an after-thought, he also put the sleeping pill in his pocket. Then he drank half a glass of water and painfully edged himself out the window. His chest scraped the ledge, and it was all he could do to strangle an out-cry of pain.

At the foot of the ladder, he staggered and nearly fell. But after a moment's rest, he squared his shoulders and walked across a corner of the lawn, into the shadows and the night.

The Los Angeles Mirror-News got further than any other paper with the story of Dr. John O'Hara Smith's mysterious disappearance from General Hospital, leaving behind a bed riddled with three bullets. In fact, the Mirror-News story had cleared the copy desk and was on its way down to the composing room before it was killed by the managing editor "for security reasons".

An all-points police bulletin was sent out, but no one was optimistic about immediate results. When you can't admit a man is missing, when you can't publish his photograph, you deprive yourself of the eyes and ears of the public, which turn up seventy-five percent of the leads in missing persons cases.

Security considerations posed three alternatives:

If Dr. Smith was telling the truth, then it was better to let whoever had twice tried to kill him wonder whether the second attempt had been successful.

If Smith had broken with an espionage ring, and had been marked for death by former associates, the various agencies concerned with security wanted a chance to find him first.

If Smith was playing some devious game of his own, let him make the next move.

As days went by, telephone circuits from Washington to Los Angeles carried messages that grew increasingly uncomplimentary. FBI headquarters hinted that certain field representatives might be transferred from Southern California to southern Kansas if results in the Smith case were not forthcoming promptly. The Air Force suggested that if both Dr. Smith and the X-15 prototype continued to be among the missing, it would not be wise to present the pending promotion of General Sanders to the White House.

The General was moodily digesting this thought, while half-listening to a discussion at a morning staff conference, when an aide whispered:

"A call from the North American Lancaster plant, Sir. It's urgent—and personal...."

General Sanders excused himself and hurried into his adjoining private office.

"Sanders," he barked.

The high, imperious voice that replied was instantly recognizable:

"General Sanders, I suggest you don't try to have this call traced, or we might not be able to finish our conversation!"

The General pressed his intercom button and held the connection open, waiting for a chance to use it.

"Go ahead, Smith," he said.

"I'll come directly to the point," said Smith. "I want two things: A place to work in safety and the funds to build another Eddie!"

"And what makes you think you can get them from me?"

"Because Eddie can help you find the X-15."

The General hunched closer to the intercom, raising his voice.

"Smith," he stalled, "why don't you come in and talk things over?"

"I do not intend to sit around waiting to be killed while your security bunglers try to decide whether I'm telling the truth!"

A Staff Sergeant looked in the door.

"Is anything wrong, Sir?"

The General motioned for silence, then scrawled on a note pad:

"Trace this call!"

"Now, Dr. Smith," he said, "if you're telling us the truth, you've got nothing to worry about...."

"General," Smith replied acidly, "do you know any better way of convincing you than to let Eddie find the X-15?"

"Well, I—"

"Goodbye, General. You think it over—and I'll call you later. Your word will be sufficient!"

The phone clicked, and General Sanders cursed bitterly. Later, he talked it over with Amos Busch, who nodded agreement to the General's proposal.

"Sure," he said. "It's worth a gamble—and we'll have Smith where we want him!"

When John O'Hara Smith phoned that afternoon, the General said promptly:

"Come on in, Dr. Smith—you've got a deal."

The available records on this phase of the case show that a Dr. J. O. Smith and three "assistants" were added to the payroll of a small Pasadena electronics firm on September 17, 1955. They were installed in one wing on the top floor of the building. The entrance to this wing was sealed off with the familiar sign: "Restricted—Permission to enter granted only on a need-to-know basis".

Apparently, few needed to know, for Smith and his assistants seldom had visitors. Deliveries of electronics components were received by one of the assistants. The four men arrived together, and left together. They brought their lunch.

Dr. Smith, of course, had been interrogated briefly when he had turned himself in at USAF Western Division Headquarters. But only the General and Amos Busch had questioned him this time.

"Look, Smith," said Amos, "if we're supposed to protect you, I want to know from what—and why it's necessary...."

John O'Hara Smith looked almost embarrassed.

"I suppose I made the same error that is so often made in declassifying information...."

"How's that?"

"When information is declassified, it's done without mathematically computing the infinite number of possible ways such information may be

useful to a hostile government.... Of course, you need an Eddie to make such a computation!"

"What's this got to do with trying to knock you off?" Busch demanded.

"It's quite evident that someone read my article in the Research Engineers' journal more carefully than you did! As a matter of fact, Eddie actually warned me that anyone hostile to the United States could not possibly allow my work to continue!"

Amos Busch and General Sanders exchanged wary glances.

"All right," said General Sanders, "We'll let that go for the moment—but what made you ask about the X-15 in the first place?"

"Eddie suggested that if the ICBM missiles could theoretically be stolen over the mid-Atlantic, it would be vastly less difficult to steal an X-15 over the Mojave Desert!"

As the two Air Force men digested this statement, along with the indisputable fact that an X-15 *had* disappeared, John O'Hara Smith blandly informed them:

"Incidentally, gentlemen, you'll have to get Eddie's duplicate tapes for me."

Busch reddened, and could not resist asking:

"Including those short-wave broadcasts from Moscow Radio?"

"Naturally!" Dr. Smith snapped. "I'm sure Eddie extracts a great deal of useful information from them!"

This second interrogation, like the previous one in the hospital, ended on a triumphant note for the exasperating Dr. Smith. When they were alone, General Sanders turned to Busch and sighed:

"We've got a double security problem, Amos! If word of this deal with Smith gets back to Washington, I'll be laughed right out of the service!"

But the General didn't begin to grasp the full implications of his predicament until the afternoon of Oct. 7, when Dr. Smith phoned to say Eddie was completed.

"Good," grunted the General. "Get going, then!"

"We'll need more information first."

"What kind of information?" General Sanders demanded suspiciously.

His suspicions were reinforced by Smith's terse dictum:

"Eddie must have all the facts on the X-15."

"Impossible!"

Dr. Smith's sniff indicated he nurtured utter disbelief in the concept of the impossible.

"Eddie operates on facts," he reminded the General.

General Sanders didn't sleep much that night. Neither did Amos Busch. They talked and argued until three in the morning, when the General poured one last drink and raised his glass.

"O.K.," he said grimly. "I've gone this far and I've got to go the rest of the way!"

They drank, and he continued:

"At least, now I won't have to worry about being laughed out of the service—I'll get court-martialed out!"

He jabbed viciously at an ice-cube with his forefinger.

"But there's one thing I'll do first," he promised.

"What's that, Sir?"

"Strangle Smith with my bare hands!"

General Sanders sat on a metal folding chair in front of Eddie, the educatable computer, and stared belligerently at the roughly-finished aluminum facade.

Eddie didn't look like much—certainly nothing like \$13,456.12 worth of components paid for out of the General's contingency fund. Speed had been the primary consideration in rebuilding Eddie. The exterior case was unpainted, and rather inexpertly held together with metal screws. There

were no knobs on the front panel controls. The vocoder grill was open; the input microphone simply rested on the workbench beside the case. The entire assembly measured about three feet long, two feet deep and eighteen inches high.

"O.K., what do I do now?" rasped the General.

"Just start talking—into the mike."

General Sanders took a sheaf of papers from his briefcase. He glared at Smith:

"You get the hell out of here! This is classified information!"

Dr. Smith smiled mockingly. On his way out of the room, he paused.

"The circuits will stay open—take as long as you wish."

Feeling like a combination of fool and Benedict Arnold, General Sanders cleared his throat and began to read:

"The North American X-15 is one of several projects now nearing the hardware stage that will take living men as well as instruments into the fourth environment of military activity, that of space.

"As soon as the satellite project completes preliminary exploration of the massive high energy spectrometer, the X-15's system should be ready to fly within two years. X-15s A, B, and C will explore 3000 mph, 50 mi. up; 4500 mph, 100 mi. up; and 6000 mph and over, 150 mi. up and out...."

General Sanders jerked open his tie. His tanned bald head was damp with sweat. He glanced around the empty workroom, set his jaw stubbornly and continued:

"Meanwhile, tests are in progress with a pilot model of X-15 to work out an entirely new vehicle system slow enough to maintain laminar flow in the boundary layer and fast enough to maintain control effectiveness at near sea-level environment. Unlike the ICBM which need only remain lethal for a few seconds, both the X-15 and its personnel must return to fly again...."

For three hours, General Sanders read steadily from his file material. During the last half hour, his voice grew husky, his throat dry and raw.

When he finished, he went to the door and shouted:

"All right, Smith.... Come in here and put this damn thing to work!"

Smith came in and informed him imperturbably:

"Not so fast, General! Eddie will still require a great deal more information."

"More? Dammit, I covered everything!"

"Everything you know about the X-15," Dr. Smith agreed, "but Eddie is now venturing into a new field and must have more than technical electronics and avionics data. He needs complete reports on the progress of the search to date, as well as the weather, topography, economy, history and current happenings in the entire peripheral area. I have built a supplemental circuit to accommodate this sort of material...."

General Sanders groaned.

"How the hell do I get into these things?"

During the next ten days, Eddie scanned microfilm on all the newspapers published since X-15's disappearance. Also marshalled before the scanner was every pertinent reference work available at public, private and university libraries in California.

At length, even John O'Hara Smith seemed satisfied. He shut off the scanner, turned on the selector mechanism and the vocader switch.

For two hours, Eddie did nothing, except hum contentedly, like a miniature washing machine. Occasionally, a weird, flickering pattern of multi-colored lights would trace across the scanning screen.

At 11:06 A.M., October 19, 1955, a flat, toneless voice came from the vocader grill:

"Laminar flow equilibrium temperature at mach 8.0, altitude 150,000 ft., of a point 10 ft. back from the leading edge is 1000 degrees Fahrenheit, assuming skin has 0.85 emissivity."

There was a small, whirring noise, and the vocader circuit clicked off.

"What the devil does that mean?" demanded the General.

"Your aerophysicists might like to know!" came back the tart reply.

At 1:34, Eddie clicked into action again:

"In flight between two planets, the theory of minimum energy orbit should be discarded in favor of acceleration at reduced speed for calculated periods of time."

"By the time we're flying between planets," General Sanders commented bitterly, "the record of my court-martial will be ancient history!"

Twenty minutes later, Eddie added:

"In the operation of small exploration vehicles, the fuel cell of the 4-H Clubs in Hanford and Bitterroot Creek will compete with the chemical energy of recombination for the prize sweet potato trophy."

Even John O'Hara Smith looked startled. But he recovered his aplomb instantly.

"Must be a circuit crossover," he explained. "No trouble to adjust it...."

While he probed into the interior of Eddie with a glass-handled screwdriver, General Sanders took out a fresh cigarette and shredded it between his fingers.

At 2:51, Eddie had this to report:

"Just as the basic physical precept of invariancy to reflection is not necessarily true, Newton's laws of motion may not always apply under certain circumstances. This would make it possible to penetrate and misdirect a navigational system based on the concept of inertial guidance."

General Sanders had been tilted back in his chair, half dozing. He bounced forward with a jar.

"What was that?"

Dr. Smith replayed this portion of the output tape.

"We talked about that at the hospital," he sternly recalled to the General. "And if the long-range missiles fired from Florida can be taken over in flight, what's to prevent their being guided to a submarine at sea?"

The General frowned in deep concentration, then relaxed and shook his head.

"Even if something like that would be possible, we've got nothing to worry about. Every missile carries a device which can be used to destroy it if the missile goes off course."

John O'Hara Smith shook his head like a teacher confronted with a pupil who was not too bright.

"Now, General, if an inertial guidance system can be penetrated, a destructor can be blocked."

"That's a mighty big if," the General shot back.

Dr. Smith smiled sardonically.

"It may not be so big when Eddie tells us what happened to the X-15!"

"When!" the General groaned. Then he came back to the problem of intercepted ICBM missiles. Half seriously, half sarcastically, he asked:

"What does Eddie think we should do about those missiles?"

"Undoubtedly there are other guidance systems that can't be broken so easily ... meanwhile, Eddie suggests booby-trapping the missiles so they'll explode when tampered with."

General Sanders closed his eyes again, and tilted back his chair. The frown between his eyes deepened.

It was six o'clock, and the early dusk was closing in on the workroom, before another statement came from Eddie. In its characteristic monotone, the educatable computer said:

"The existing developmental missile program will not be affected by the rising divorce rate in Bakersfield and Kern County."

Dr. John O'Hara Smith pursed his lips in disapproval.

"Eddie's not behaving at all well! I'm afraid that new circuit relay will take some working over...."

General Sanders climbed slowly to his feet. He picked up his hat.

"O.K., Smith," he said, "You sold me a bill of goods, and I bought it! Now I'm turning you and this whole damn mess back to the FBI! Let Cowles go crazy for awhile!"

As Frank Cowles sat in the General's office and heard what had been going on, he said mildly:

"Well, I guess you had to take the gamble."

"Thanks," said General Sanders. "I hope the Pentagon will look at it the same way—but I doubt it!"

"We've got a problem, too, General," Cowles pointed out. "When everything's said and done, there's absolutely no charge we can file against Smith."

"But he just can't walk away—not with all he—or that miserable Eddie—knows about the X-15!"

Cowles smiled faintly.

"I would imagine that Eddie now belongs to the Air Force."

"We'll break the damn thing up for scrap!"

The General's intercom buzzed. An aide's voice said apologetically:

"That Dr. Smith is calling you again, Sir."

"Tell him to go to hell!"

A few seconds later, the intercom buzzed again.

"Dr. Smith on the line, Sir—He says it's something about the X-15 missile."

General Sanders looked as though he wanted to sweep the intercom off his desk.

"Why not talk to him," Cowles suggested. "I'd like to hear this."

The General picked up his phone, and said with deceptive calm:

"All right, Smith ... make it short."

"It was the logging truck," Dr. Smith replied, in his most superior manner.

"Huh?"

"Eddie's circuit is coordinated now. He says that the same afternoon the X-15 disappeared, a passenger car ran into the back of a logging truck

northbound on Highway 395, about fifty miles from the Lancaster base. Two people were killed...."

"Smith, what kind of pipedream are you peddling now?"

"General, the truck was loaded with redwood logs and heading north!"

"I don't give a damn where it was going!"

"Wait, General!" Dr. Smith's tone was almost a command. "Eddie wants to know why a logging truck was traveling *toward* the redwood country with a load of logs. He also points out that the X-15 is about the size of a redwood log, and could be concealed perfectly in the middle of a load!"

The General seemed to be swallowing something angular and unpleasant.

"We'll check that truck," he said, at last. "But remember, Smith, you've had it—you'll never hook me again!"

He put down the phone, and said to Cowles:

"You get on the merry-go-round this time!"

The California Highway Patrol in Mojave had the report on the accident. Clearly, it had been the fault of the passenger car. The truck driver was identified in the report as Art Backus, an independent hauler, working out of Eureka, located on the far northern tip of the California coast, about eight hundred miles from the scene of the accident.

A routine check by the FBI disclosed that Backus had done time in San Quentin on a morals charge involving a minor girl. He had driven trucks for a dozen lumber companies in northwest California until the past summer, when he had bought a new truck and trailer, for cash, and gone into business for himself.

Two FBI agents stepped up to him in a roadside cafe on Highway 1, between Eureka and Trinidad Bay. A gaunt, stooped man, he nearly collapsed when the agents showed him their identifications. He was broken, and ready to talk, even before mention was made of the fact that the penalty for peace-time espionage is death.

Backus guided the FBI to an abandoned sawmill, some two miles inland, where the X-15 had been taken apart, minutely photographed, and then sunk in the old log pond.

The men who had hired Backus and dismantled the X-15 had left the area several weeks earlier. They were remembered with friendliness by the residents of Trinidad Bay, who described them as "real nice guys and good fishermen, too." They had told Backus they would be back in the late autumn for the steelhead run, and perhaps would have some more hauling business for him at that time.

The FBI offered Backus one chance for life. He accepted it, with abject eagerness.

Beyond this point, there are no more available records on the case of Dr. John O'Hara Smith, and Eddie, the educatable computer. But several items, not apparently related in any way, make interesting speculation.

On January 3, 1956, the Air Force reported that a Thor intermediate-range ballistic missile, launched from Patrick Air Force Base, Florida, had been destroyed when it appeared to be wandering off course.

About the same date, a Panamanian freighter, riding the gulf-stream toward the West Indies, radioed a report of sighting a massive oil slick and a scattering of debris, some of it bearing Russian insignia. No survivors were found.

The U.S. State Department solicitously inquired of the Soviet Union if any of its vessels had been lost in the winter storms of the Caribbean. The Soviet Union testily replied that no Soviet vessels could have been lost, since Soviet vessels, as a matter of sound international principle, confined their operations to their own territorial waters.

During Easter Week of 1956, the FBI announced the arrests of four men on charges of espionage: A druggist in Tucson, Arizona; an importer in San Francisco; a retired real-estate operator in Los Angeles; an obscure trucker in northern California. All pleaded guilty in order to escape the gas

chamber. The details of the charges against them were not disclosed, except to members of a Federal Grand Jury.

Two other published items are worth noting:

The May, 1956, issue of the journal published by the Institute of Research Engineers reported that one of its members, Dr. J. O. Smith, had recovered from injuries suffered in the explosion of a butane stove and had accepted a government research position in Washington, D.C.

The other item was a paragraph in Aviation Weekly, congratulating Major General David William Sanders on his promotion to Brigadier General.

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